13 The points A, B, C and D lie in order on a straight line such that

$AB : BD = 1 : 2$
$AC : CD = 7 : 2$

Find $AB : BC : CD$

AB : BD = 1 : 2, so AD is 3 parts.

AC : CD = 7 : 2, so AD is 9 parts.

1 mark
Match the SAME whole length AD.

Use a common total of 9 parts: AB : BD = 3 : 6.

AC = 7 parts, so BC = 7 – 3 = 4 parts.

1 mark
Subtract AB from AC to find BC.

AB : BC : CD = 3 : 4 : 2

1 mark
Check: AB:BD = 3:6 and AC:CD = 7:2.

3 : 4 : 2

(Total for Question 13 is 3 marks)

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